



Intelligent Distributed Systems
Electronics and Robotics

Final Report

**Model for managing the heating of a house
organized with autonomous and interconnected
entities**

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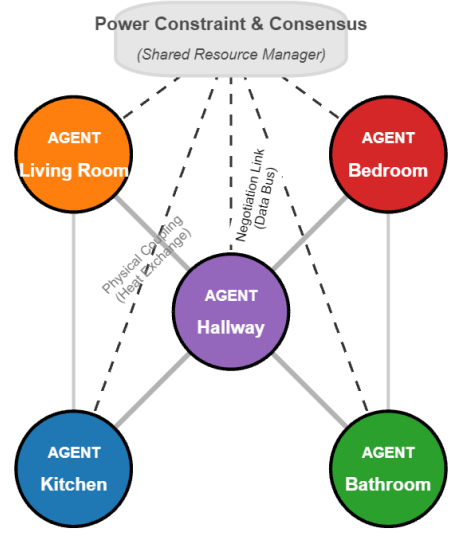
1 Abstract and Introduction

The energy transition and the growing use of renewable sources impose new challenges on the stability of the electric grid (Smart Grid). In this scenario, the residential sector, responsible for a significant share of global consumption, must evolve from a passive load to an active and flexible system. However, centralized management of complex multi-zone residential complexes presents scalability and privacy problems. This work proposes a distributed approach based on Multi-Agent Systems (MAS) for domestic energy coordination, aiming to guarantee thermal comfort while respecting stringent power constraints imposed by the grid, without the aid of a central controller. Modern homes are predisposed to smart management of their resources. The presented system is a distributed control for a multi-zone home, where each room is an autonomous agent, but coordinated with the others. Unlike classic thermostats, this system integrates real weather forecasts, wall thermodynamic inertia, and energy resource negotiation. The power peak is thus managed, keeping consumption limited, while also ensuring comfort.

2 Distributed System Adopted

As anticipated in the introductory phase, for the management of comfort and energy in the domestic environment, an architecture based on Multi-Agent Systems (MAS) has been adopted [6].

In this model (Fig. 1), the system is not controlled by a single monolithic central unit, which would represent a single point of failure and limit scalability: The system demonstrates high Scalability: adding a new room entails only a local modification to the adjacency matrix, without requiring the reconfiguration of the entire global controller. The system is instead decomposed into autonomous and cooperative entities. Specifically, each "Room" is modeled as an independent Intelligent Agent.



Network Topology: Interconnected Multi-Agent System

Fig. 1: Distributed System Scheme

To summarize the general operation, the characteristics that each room possesses are:

- **Local and Peripheral Perception:** Monitors its own state variables (internal temperature, user presence) via dedicated sensors, without direct access to the raw data of other agents.
- **Decision Autonomy:** Calculates its own ideal thermal requirement based on local objectives (comfort setpoint) and internal dynamics.
- **Actuation:** Controls its own devices (heating valves, window actuators).

The true distributed system aspect emerges in the management of shared resources. Since the total power available from the grid is limited to a threshold P_{Grid}^{max} (fixed at 3000 W), the agents cannot act in total isolation. If the sum of local requests exceeds availability, the system activates a distributed coordination mechanism.

The agents "negotiate" access to energy or self-regulate based on dynamic priorities, ensuring that the global constraint (Hard Constraint) is respected without the need for a supervisor micro-managing every valve [2]. This approach allows the system to be robust and modular: adding a new room does not require rewriting the entire control

code, but only adding a new agent to the cluster.

2.1 Resource Allocation and Priority Management Algorithm

To manage the strict constraint imposed by the grid (P_{grid}^{max}), the system implements a competitive coordination mechanism based on priorities [3, 5]. Unlike purely iterative approaches, this module acts as a Resource Allocator that resolves the resource conflict at each simulation step (dt). The algorithm receives two vectors as input from the N agents: the ideal power request vector (P_{demand}) and the priority vector (ρ), the latter dynamically derived from the probability of user presence in the room. The allocation procedure follows three logical phases:

1. Data Aggregation: The system collects the tuples $(ID_i, P_{demand,i}, \rho_i)$ for every room i .
2. Priority Sorting: Agents are sorted in descending order based on their priority. In this way, occupied rooms (high probability of presence) obtain the right of preemption on the energy resource.
3. Greedy Allocation: The system iterates over the sorted list, distributing the available power $P_{residua}$ (initially equal to P_{grid}^{max}).

$$P_{grant,i} = \min(P_{demand,i}, P_{residua})$$

Successively, the residual power is updated:

$$P_{residua} \leftarrow P_{residua} - P_{grant,i}$$

This approach guarantees that the Hard Constraint is always respected ($\sum P_{grant} \leq P_{grid}^{max}$), temporarily sacrificing the comfort of low-priority rooms (which receive 0W if the budget is exhausted) in favor of critical ones.

Note: The proof of the algorithm is given in the appendix A.

2.2 Local Agent Intelligence (Distributed Brain)

Each thermal zone is governed by a local controller ("Distributed Brain") that operates autonomously, calculating the ideal energy requirement before submitting it to global negotiation. The local control algorithm integrates various logics to maximize efficiency.

The core of the control is a Proportional-Integral-Derivative (PID) regulator that calculates the power necessary to eliminate the error between the measured temperature (T_{meas}) and the setpoint (T_{set}). A saturation mechanism (clamping) is implemented on the integral action to avoid the windup phenomenon when the actuator is saturated.

The agent monitors boundary conditions. If window opening is detected (via comparison between T_{meas} and external temperature T_{ext} with hysteresis), heating is inhibited to avoid waste. Furthermore, the setpoint is dynamically adapted based on weather forecasts (T_{future}) to prevent pipe freezing (Anti-freeze logic).

To avoid sudden absorption peaks (spikes) harmful to the electric grid, the physical actuation of the granted power (P_{hvac}) is filtered by a limiting ramp (2.0W/s per step), which ensures fluid energy transitions. The output of this block is twofold: the Power Demand (how much the room would like to consume) and the Priority Index (how urgent this request is based on human presence).

3 System Model

This section describes the "Digital Twin" model developed to realistically simulate the thermal dynamics of the residential building. The model was implemented in the MATLAB/Simulink environment.

3.1 Thermal Zoning and Topology

The simulated environment is subdivided into 5 independent thermal zones (Kitchen, Living Room, Bedroom, Bathroom, Hallway), each characterized by heterogeneous physical properties. Unlike simplified models, each

room possesses specific thermal capacity and insulation parameters:

- **Living Room:** High dispersion zone (large windows) and large volume ($C_{aria} = 250\text{kJ/K}$).
- **Bathroom:** Low inertia zone but with high comfort demand.
- **Bedroom and Kitchen:** Medium inertia zones.
- **Hallway:** Zone with low comfort demand, but highly influenced by the 4 adjacent rooms.

Each agent is composed of three sub-blocks: a cluster for temperature sensors, a physics implementation for heat propagation, and its own "distributed brain" that reactively controls the heating mechanism associated with that room (PID).

The rooms take various values as input and produce others to create a dense network of information, necessary so that each unit can manage its own heating, having known the conditions of the other entities composing the system. In particular, each room takes 7 inputs: target temperature, external temperature, total disturbance power, average neighbor temperature, future temperature based on weather prediction, power granted by the central power distributor, and the probability of having a person in the room.

Note: the final occupancy signal is not stochastic, but deterministic and is obtained from the logical union of different inputs, thus returning only values equal to 0 and 1.

The outputs produced are 4: actual internal temperature, power requested by the room itself, heating priority determined by the occupancy status and consumed power.

The rooms are not thermally isolated from each other but exchange heat through partition walls, creating a strongly coupled system where the thermal state of one room influences the adjacent ones.

3.2 Two-Mass Thermal Model

To model the thermal inertia necessary for validating the distributed control algorithm, a

second-order lumped parameter model (Two-Mass Model) [4] was adopted. The dynamics of each room i is described by two differential equations coupled via thermal resistances (R), defining the following state variables:

- **Air Mass (T_{air}):** Represents the internal temperature. Having a reduced thermal capacity (C_{aria}) it presents fast dynamics. It is the node upon which the control input (P_{hvac}) and disturbances act directly, making the air temperature rapidly reactive to system decisions.
- **Wall Mass (T_{wall}):** Represents the building structure. With a high thermal capacity (C_{muri}) (e.g., 60MJ/K), it acts as a "thermal battery" and a low-pass filter towards the external temperature (T_{ext}), stabilizing the system.

The resulting mathematical model is described by the following system:

$$\begin{cases} C_{aria} \cdot \dot{T}_{aria} = P_{hvac} + P_{dist} - \frac{T_{aria} - T_{wall}}{R_{ia}} - \frac{T_{aria} - T_{vicini}}{R_{int}} \\ C_{muri} \cdot \dot{T}_{muri} = \frac{T_{aria} - T_{wall}}{R_{ia}} - \frac{T_{wall} - T_{ext}}{R_{ext}} \end{cases}$$

3.3 Environmental Input and Disturbances

The model does not operate in a vacuum but is subject to exogenous and endogenous disturbances that test the control system.

First, the system is integrated with the Open-Meteo API [1] to acquire real external temperature (T_{ext}) and solar irradiation (Watt/m^2) profiles based on the geographic coordinates of Trento (Lat 46.07, Lon 11.12). Solar gain (Q_{sun}) is calculated dynamically and influences rooms based on the glazed surface area ($Area = 2.0\text{m}^2$, $\text{SHGC} = 0.7$).

Furthermore, to simulate the unpredictability of the use of domestic appliances, specific heat disturbances have been modeled for each room, which occur randomly (Tab. 1).

Room	Activity	Power
Kitchen	Stove	+1500W
Bathroom	Shower	+800W
Living Room	People Presence	+450W
Bedroom	People Presence	+160W

Tab. 1: Additional Contributions for heat supply

3.4 People Presence Probability Management

The management of the presence or absence of people in the room is based on a logical model. Having to simulate this aspect, the most suitable approach was to handle presence as a stochastic variable, determined by a probability distribution (Fig. 2). All this was implemented via a lookup table with a specific probability vector for each room. Connecting to a clock, it generates a time-dependent probability profile (e.g., Gaussian-shaped) representing the likelihood of presence of people in the room in continuous time. The probability vectors are arbitrarily decided based on the activities usually associated with that room for a specific time.

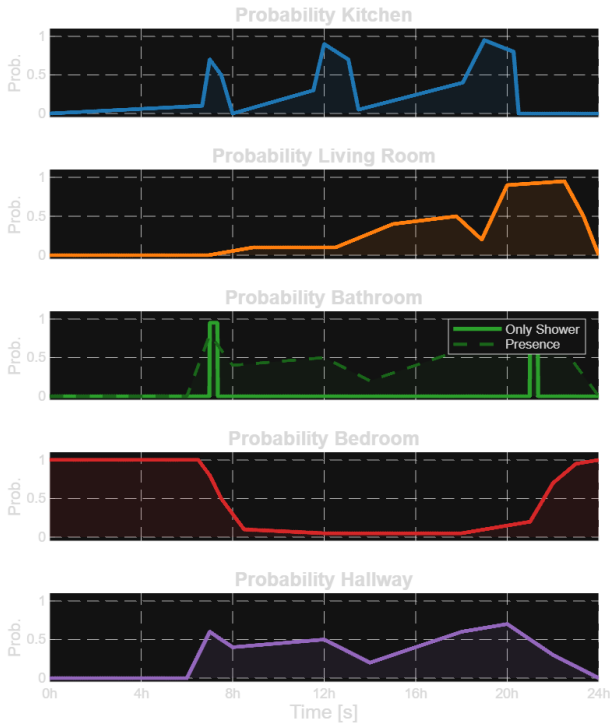


Fig. 2: Probability distribution for people presence

A random generator between 0 and 1 is compared with the value extracted for the probability at a certain instant in time. If the random value is lower than the probabilistic one, the switch goes to off: there is no one in the room, so the requested temperature target is lower. In case the opposite occurs, the switch goes to on and the set point is higher. Set points vary from room to room, but the fact that the one for the empty room is lower than the one for the occupied room

is a general characteristic.

To better manage consumption, we decided to place a double check to compare the logical output regarding presence in the room between two consecutive iterations (every 5 minutes). This is done by evaluating the previous result with the current one using an AND logic. If both register presence, heating effectively activates. This serves to manage both short stays that do not require heating activation and potential false positives.

3.5 Sensing and Actuation Constraints

Each zone is equipped with a sensor array (modeled with Gaussian noise $\sigma = 0.05$) for temperature monitoring.

The actuation system is bound by precise physical limits. Each Fan-Coil has a maximum saturation equal to $P_{max} = 1200W$. Furthermore, the general grid constraint (Global Constraint) is recalled again: The entire apartment cannot absorb more than $P_{grid}^{max} = 3000W$ instantaneously from the grid. This physical constraint is the main driver making the distributed approach described in the previous section necessary.

4 Simulation Results Analysis

To validate the effectiveness of the distributed control system, the results of an extended simulation over a time horizon of three days (255.600 seconds) with a fixed integration step are reported. (The required computational time was approximately 1h 30m, confirming the feasibility of the approach even on non-dedicated hardware).

4.1 Temperature Trend

The graph reported in Fig. 3 illustrates the temporal evolution of temperatures in the five thermal zones.

The analysis of the results highlights three phases or characteristic behaviors:

- **Power-on Transient (Cold Start):**
At instant $t = 0$, the system is initialized in critical conditions, with internal

temperature equal to the external one obtained from weather data ($T_{ext} = 4.3^\circ C$). A rapid rise in temperatures is observed in all rooms. In this phase, the consensus algorithm is under maximum stress since the aggregate request amply exceeds the grid constraint P_{grid}^{max} . The system manages saturation by distributing available power, allowing rooms to reach the operational comfort band in reasonable times without tripping the meter.

- **Presence-based Modulation (Setpoint Tracking):** Once at steady state, the visible oscillations are not instabilities, but the correct response to the dynamic setpoint change. When the stochastic model detects (or predicts) the presence of people, the temperature target rises ("Comfort" phase, higher peaks); when the room empties, the target lowers ("Eco" phase). The graph clearly shows how each agent tracks its own local reference independently but coordinately.

- **Inertia and Coupling:** It is noted how temperature drops (cooling phases) are much slower than rises. This validates the effectiveness of the two-mass model: the thermal capacity of walls (C_{muri}) releases heat gradually, stabilizing the environment. Furthermore, the trend of the "Hallway" (orange/low line) appears smoother compared to other rooms: acting as a transit zone with lower setpoints and thermal coupling with all other zones, it effectively acts as a thermal "buffer" for the entire apartment.

Furthermore, the 'step' trend visible in the main rooms (e.g., Living Room, green line) confirms the correct system response to stochastic stimuli: temperature is maintained at the comfort level only when presence is detected (or predicted), reducing to the 'Eco' level during inactivity periods to maximize energy savings.

A detailed analysis of the individual thermal traces allows appreciating the correlation between heating activation and room

usage typologies. The service zones show impulsive dynamics: the Kitchen (purple line) presents steep and localized peaks corresponding to meals (e.g., $t \approx 43.200s$), where the controller action is amplified by stove heat (+1500W), while the Bathroom (blue line) highlights rapid oscillations typical of short but intense sanitary use (e.g., shower).

Opposite in nature is the behavior of residential zones: Living Room (green line) and Bedroom (yellow line) exhibit 'extended occupation' profiles, characterized by wider and more regular temperature plateaus. This reflects prolonged stay windows (such as evening relaxation or night rest), where the system must sustain comfort for consecutive hours rather than for brief moments.

Finally, acting as a thermal binder, we find the Hallway (orange line): its stability, devoid of endogenous disturbances and with dampened oscillations, confirms its role as a passive buffer integrating thermal flows coming from adjacent rooms.

4.2 Energy Consumption

From the analysis of cumulative energy consumption, reported in Fig. 4, a monotonically increasing trend typical of power integration over time is observed. The curve allows distinctly distinguishing two operating regimes by analyzing the variation of its slope (derivative).

The first Transient phase (Energy Investment) corresponds to the initial segment where a very marked slope is noted. This rapid growth is attributable to the Cold Start condition: starting from an initial temperature (T_{start}) equal to the external one ($4.3^\circ C$), the system must provide a large amount of energy in a short time to overcome the structure's thermal inertia and bring air and wall masses to steady-state temperature. This initial "energy investment" significantly affects the total balance.

The Steady-State phase (Maintenance), instead, is observed after overcoming the initial transient, where the curve assumes an almost linear trend with reduced and constant slope. This indicates that, once the setpoint is reached, the system requires low

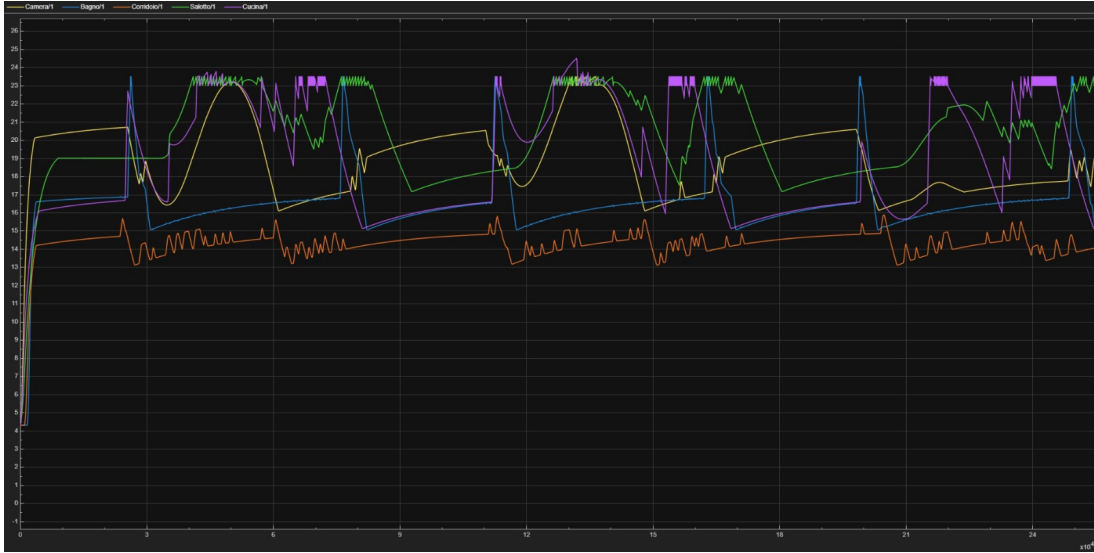


Fig. 3: Temperature Trend

and stable average power, necessary exclusively to compensate for thermal dispersions towards the outside. The linearity of this phase demonstrates the excellent stability of the control algorithm, which avoids waste due to excessive fluctuations.

In quantitative terms, the total energy consumption over the three simulation days stands at approximately $25kWh$, corresponding to a daily average of $8.3kWh$. This value results extremely competitive when compared with average consumption for winter heating of equivalent housing units (often exceeding $10 - 15kWh$ daily in the absence of smart optimization). Finally, note that the result is conservative: in a real scenario, where internal temperature never drops to the drastic external values of the initial condition, energy consumption for maintenance alone would result even lower.

Finally, it is fundamental to underline the effectiveness of the algorithm in terms of Peak Shaving. In an uncontrolled scenario, the simultaneous activation of all zones (especially during Cold Start) would entail a theoretical instantaneous power demand equal to the sum of nominal loads ($\sum P_{max,i} \approx 6.0kW$), well beyond domestic contractual limits. The distributed system, instead, forcibly saturated the request to $3.0kW$, flattening the load curve. This

improvement in Load Factor (ratio between average and peak load) confirms that the proposed architecture not only optimizes local comfort but contributes actively to electric grid stability (Grid Stability).

The results of some recent simulations are reported to demonstrate the validity of the system (Tab. 2):

Scenario	Start T_{ext}	Weather	Total Energy	Daily Avg
Case A (Reported)	$4.3^{\circ}C$	Mixed/Cloudy	$25.1 kWh$	$8.37 kWh$
Case B (Sunny Days)	$4.5^{\circ}C$	High Irradiance	$22.8 kWh$	$7.60 kWh$
Case C (Cold Wave)	$-1.2^{\circ}C$	Overcast	$31.5 kWh$	$10.5 kWh$
Case D (Normal Day)	$6^{\circ}C$	Mixed/Cloudy	$23.4 kWh$	$7.80 kWh$

Tab. 2: Comparison of Energy Consumption under different weather scenarios (3-day horizon)

5 Conclusions and Future Directions

The simulations confirm the operational validity and robustness of the proposed Multi-Agent System (MAS). Unlike monolithic centralized controllers, this distributed architecture enhances fault tolerance (avoiding *Single Points of Failure*) and ensures locality. A decisive advantage is scalability: the system supports an arbitrary number of thermal zones (N), where adding a new room simply requires instantiating a local agent without global reconfiguration.

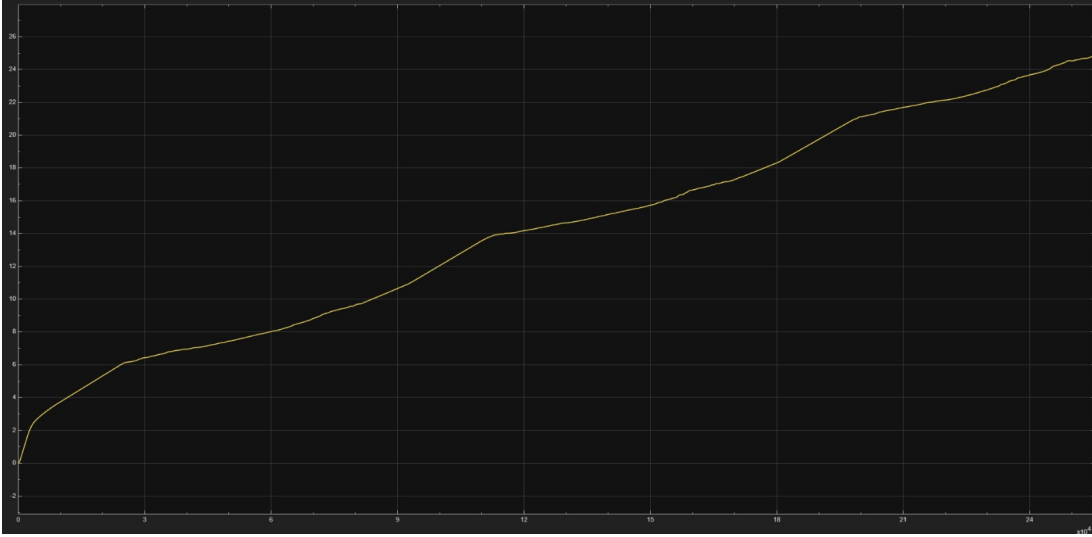


Fig. 4: Evolution of Power Consumption

Energy consumption results (≈ 25 kWh) are conservative, primarily driven by the “Cold Start” initialization ($T_{start} = T_{ext}$); steady-state maintenance would require significantly less energy. Furthermore, the system acts as an active Demand Response node, effectively flattening power peaks (*Peak Shaving*) to strictly adhere to the 3 kW grid constraint. From an implementation perspective, the algorithm’s low computational cost ensures feasibility on low-cost embedded hardware.

Future developments could extend the model to summer cooling and integrate household appliances, managing them by distinguishing between *non-shiftable* (high priority) and *shiftable* loads (negotiated activation). Finally, integrating people-counting sensors (e.g., IR) would allow for occupancy-weighted prioritization ($Priority \propto N_{people}$). This feature was not implemented in the current study as simulating realistic stochastic multi-user flows exceeded the scope of the developed thermal model.

In conclusion, the presented system has proven to be reliable and effective in meeting heating requirements. While open to further refinements, the proposed model represents a promising alternative to traditional, obsolete heating infrastructures for better energy management.

Appendix A: Proof of Power Constraint Satisfaction

To validate the effectiveness of the power management algorithm, a mathematical proof was developed. The objective is to ensure that the aggregate granted power remains strictly below the maximum available capacity (P_{grid}^{max}). While the general framework is inspired by the stability analysis proposed in the article [7], the proof itself follows a more direct approach based on a telescoping series, demonstrating its inherent convergence to a safe operating point.

Theorem

Considering in general an N -set of agents $\mathcal{N} = \{1, \dots, N\}$. Given a demand vector P_{demand} and a maximum grid capacity $P_{grid}^{max} > 0$, the algorithm ensures that the total granted power $P_{granted}$ always satisfies the hard constraint:

$$\sum_{i=1}^N P_{granted,i} \leq P_{grid}^{max} \quad (\text{Eq. 1})$$

Proof

The logic of the algorithm is iterative and operates on a priority-sorted list of agents. Let $\sigma(k)$ represent the index of the agent with the k -th highest priority and $k = 1 \dots N$. We define $P_{rem}^{(k)}$ as the residual power remaining

after the allocation to the k -th agent, starting with the initial condition:

$$P_{rem}^{(0)} = P_{grid}^{max} \quad (\text{Eq. 2})$$

For each iteration (agent) k the granted power $g_k = P_{granted,\sigma(k)}$ is computed as follows:

$$g_k = \min(P_{demand,\sigma(k)}, P_{rem}^{(k-1)}) \quad (\text{Eq. 3})$$

The residual power is immediately updated for the subsequent step:

$$P_{rem}^{(k)} = P_{rem}^{(k-1)} - g_k \quad (\text{Eq. 4})$$

By definition (Eq. 3), the granted power g_k cannot exceed the available residual power $P_{rem}^{(k-1)}$. This implies that the residual power is bounded below by zero:

$$P_{rem}^{(k)} \geq 0 \quad \forall k \quad (\text{Eq. 5})$$

The total granted power corresponds to the sum of the individual allocations over all N steps:

$$P_{total} = \sum_{k=1}^N g_k \quad (\text{Eq. 6})$$

By substituting g_k using (Eq. 4), the summation takes this form:

$$\begin{aligned} P_{total} &= \sum_{k=1}^N (P_{rem}^{(k-1)} - P_{rem}^{(k)}) \\ &= (P_{rem}^{(0)} - P_{rem}^{(1)}) + (P_{rem}^{(1)} - P_{rem}^{(2)}) + \\ &\quad + \dots + (P_{rem}^{(N-1)} - P_{rem}^{(N)}) \\ &= P_{rem}^{(0)} - P_{rem}^{(N)} \end{aligned} \quad (\text{Eq. 7})$$

Finally, substituting the I.C. (Eq. 2)

$$P_{total} = P_{grid}^{max} - P_{rem}^{(N)} \quad (\text{Eq. 8})$$

Since we established that the final residual power is non-negative (Eq. 5), it strictly follows that:

$$P_{total} \leq P_{grid}^{max} \quad (\text{Eq. 9})$$

and so the theorem (Eq. 1) is proved. ■

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